NAME: Aahil Mukadam
DEPARTMENT: Manufacturing Systems (ESCO)
LEAD: Derek Huynh

Student Background

Education: Bachelors of Applied Science in Mechatronics Engineering, University of Waterloo (Sep 2022 - Apr 2027)

Work Experience:

1. Engineering Analyst, TMMC (Sep 2025 - Dec 2025)
2. Automotive Technician, Paradigm Auto (Jan 2025 - Apr 2025)
3. T.A., University of Waterloo (Sep 2023 - Dec 2023)
4. Support Analyst, Building Innovation (Jan 2023 - Apr 2023)

Personal Objectives

1. Learn key manufacturing engineering concepts to gain a clearer understanding of potential methods for energy savings
2. Work with innovative technologies, such as VTScada, to explore applications in energy efficiency
3. Familiarize myself with all shop areas and their processes to accurately and efficiently apply energy-saving principles

Schedule of Projects

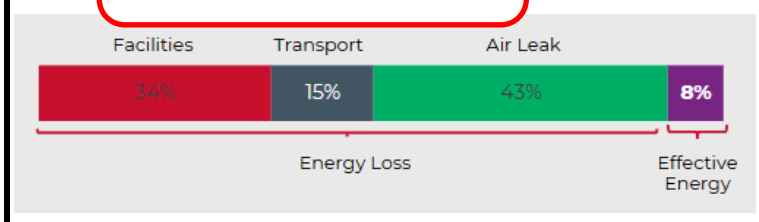
Planned / Actual

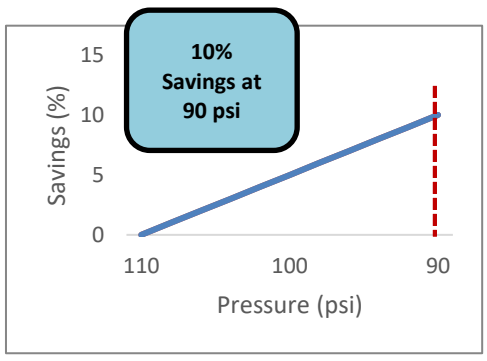
Project	Months	September					October				November				December				Status
	Week	1	8	15	22	29	6	13	20	27	3	10	17	24	1	8	15	22	
West Weld Shell Body Pneumatics																			◕
West Weld Shell Body Booster																			◑
VTScada Cambridge Release																			◕
West Assembly CA Mapping																			●

STEP 1. Clarify the Problem (must be measurable and units consistent throughout)

Background: Facilities attempted to reduce pressure to **90 psi** throughout TMMC West, as a part of Toyota's energy savings initiative.

Compressed Air is the **least** efficient source of energy, only utilising **8%** of intial energy due to **various** losses



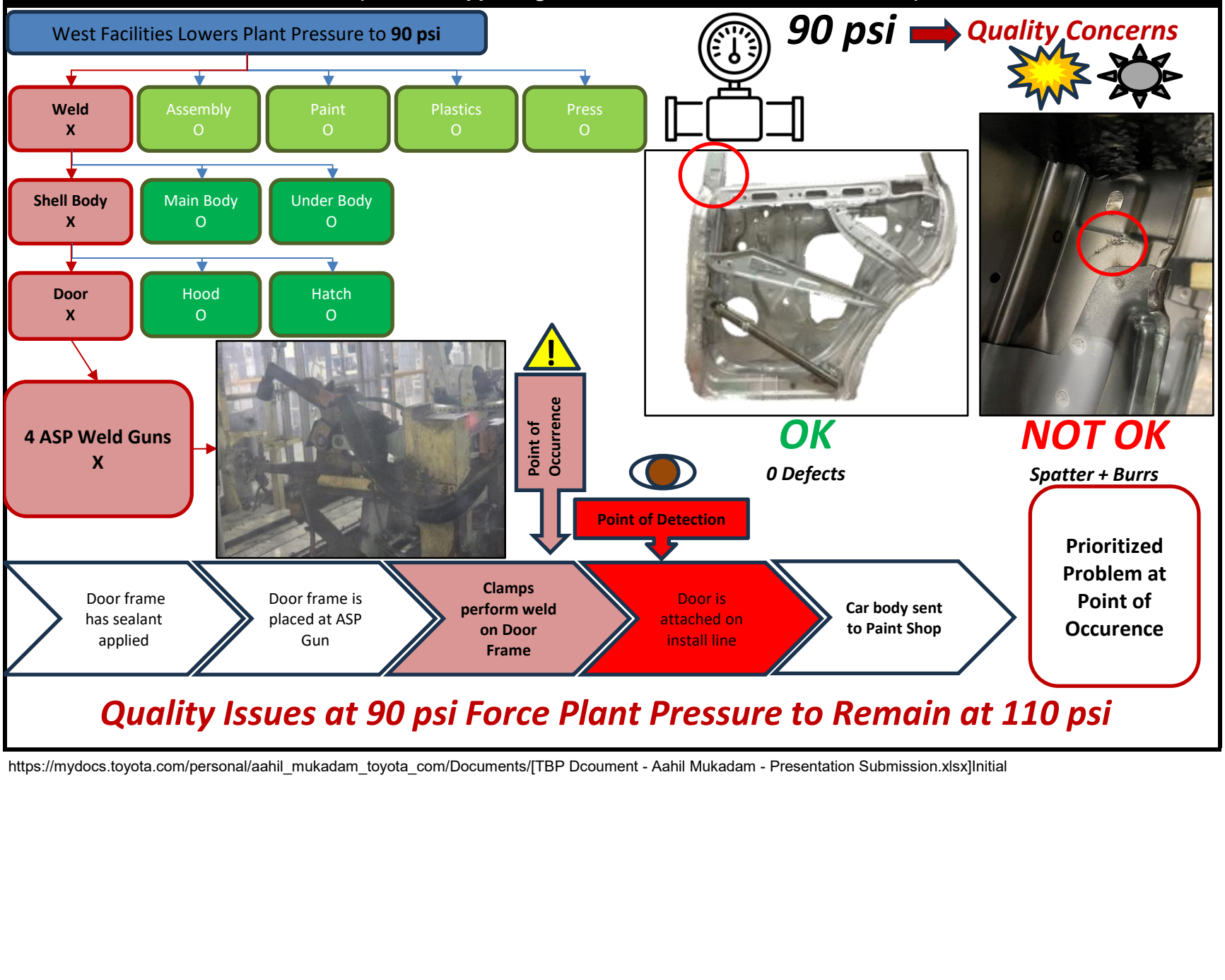


IDEAL 90 Psi
GAP
Current 110 Psi

20 Psi
($116,000/year)
(86.45 tCO2/year)
(10,080,690 m^3)

ULTIMATE GOAL
Enable compressed air pressure reduction without impacting shop KPI's

STEP 2: Breakdown the Problem (what is happening, when, who, how, where.? NOT WHY)



Quality Issues at 90 psi Force Plant Pressure to Remain at 110 psi

https://mydocs.toyota.com/personal/aahil_mukadam_toyota_com/Documents/[TBP Dcoument - Aahil Mukadam - Presentation Submission.xlsx]Initial

	Student	Student Lead	Manager	Sr. Mgr
		Derek Huynh (Dec 9, 2025 15:41:20 EST)	Mark Groen	Chris Maltby (Dec 10, 2025 08:16:46 EST)

STEP 3: Target (SMART)

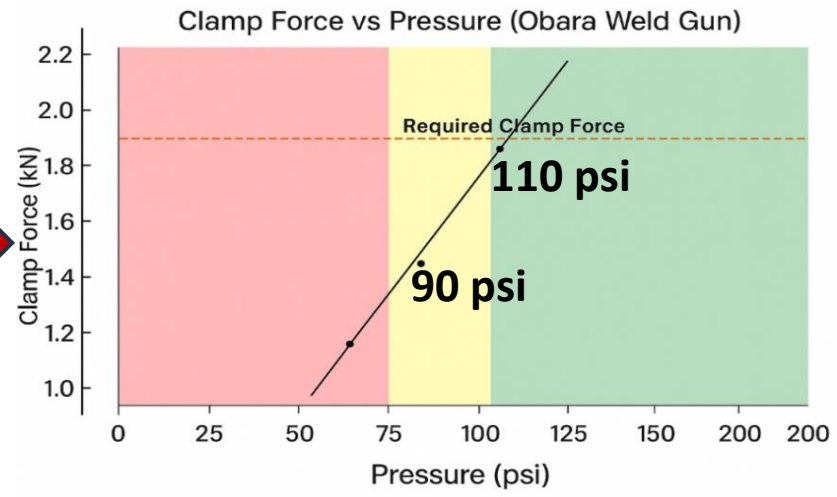
ASP Weld Gun Runs at 100 psi with 0 Quality issues by the end of 2026

STEP 4. Root Cause Analysis (Narrow down to 1 root cause)

Weld Gun at 90 psi
↓
Quality Issues
↓
Overheating
↓
High Weld Resistance
↓
Insufficient Clamp Force

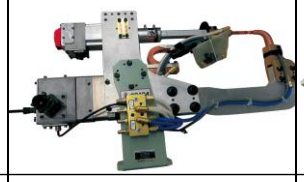
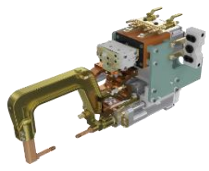


Required Clamp Force (1.93kN)



Clamp Force Can Not be Achieved Below 110 psi

STEP 5/6. Develop and Implement Countermeasures (Post SOP Launch)

				
Measure	Electric Weld Gun	Local Booster	Line Size Up	Cylinder Size Up
Clamp Force	O - 2.0kN	O - 2.0kN	O - 2.0kN	O - ≈2.0kN
Complexity	X - New Procedures	O/Δ - Tank Refill period	O - Procedures Unchanged	O - Procedures Unchanged
Cost	X - ≈60,000 USD	O - ≈4,000 USD	Δ - ≈18,000 USD	O - ≈3,700 USD
Maintainability	O - Minimal Maintenance	Δ - Leaks, Consistent Upkeep	O - Minimal Maintenance	O - Higher Demand
Outcome	**X**	**Δ**	**Δ**	**O**



STEP 7. Results and Process Evaluation
(Before vs after condition, against target)

Projected Outcome

Process Runs at 100 psi

0 Quality Issues

Step 8. Standardization and Yokoten
(What, where, when,who, how)

2026 and Beyond

Identify Pneumatics in West Weld

TMMC Plant Pressure Reduction

Analysis of Personal Objectives	Eval.
1. Learn key manufacturing engineering concepts to gain a clearer understanding of potential methods for energy savings	O
2. Work with innovative technologies, such as VTScada, to explore applications in energy efficiency	O
3. Familiarize myself with all shop areas and their processes to accurately and efficiently apply energy-saving principles	O

Reflections/Recommendation for Co-op Program

As a member of team, that consistently works with a variety of different shops, giving students the opportunity to work on projects with/related to other shops could be beneficial to growth and understanding of TMMC processes.